

## SAFETY DATA SHEET

Revision Date 02-April-2024

Revision Number 3

### 1. Identification

**Product Name** Direct Methanol Fuel Cell (DMFC) Cathode, 50cm3

**Cat No. :** 45368

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

**Carcinogenicity** Category 2

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Suspected of causing cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use

Do not handle until all safety precautions have been read and understood

Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF exposed or concerned: Get medical advice/attention

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component               | CAS-No    | Weight % |
|-------------------------|-----------|----------|
| Graphite                | 7782-42-5 | 60       |
| Polytetrafluoroethylene | 9002-84-0 | 5        |
| Carbon black            | 1333-86-4 | 5        |

### 4. First-aid measures

|                                                               |                                                                                                                         |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                                             | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Ingestion</b>                                              | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | None reasonably foreseeable.<br>Treat symptomatically                                                                   |

### 5. Fire-fighting measures

|                                       |                                                                                |
|---------------------------------------|--------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Dry sand. Carbon dioxide (CO <sub>2</sub> ). Powder. Do not use water or foam. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                       |
| <b>Flash Point</b>                    | No information available                                                       |
| <b>Method -</b>                       | No information available                                                       |
| <b>Autoignition Temperature</b>       | No information available                                                       |
| <b>Explosion Limits</b>               |                                                                                |
| <b>Upper</b>                          | No data available                                                              |
| <b>Lower</b>                          | No data available                                                              |

**Sensitivity to Mechanical Impact** No information available  
**Sensitivity to Static Discharge** No information available

#### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Hydrogen fluoride. Metal oxides.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
1

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
-

### 6. Accidental release measures

#### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

#### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### 7. Handling and storage

#### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

#### Storage.

Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Organic materials. Oxidizing agent.

### 8. Exposure controls / personal protection

#### Exposure Guidelines

| Component               | Alberta                    | British Columbia         | Ontario TWA/EV           | Quebec                   | ACGIH TLV                | OSHA PEL                                                                                                                                                                   | NIOSH                                                                                    |
|-------------------------|----------------------------|--------------------------|--------------------------|--------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Graphite                | TWA: 2 mg/m <sup>3</sup>   | TWA: 2 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> | (Vacated) TWA: 2.5 mg/m <sup>3</sup><br>(Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> | IDLH: 1250 mg/m <sup>3</sup><br>TWA: 2.5 mg/m <sup>3</sup>                               |
| Polytetrafluoroethylene |                            |                          |                          | TWA:                     |                          |                                                                                                                                                                            |                                                                                          |
| Carbon black            | TWA: 3.5 mg/m <sup>3</sup> | TWA: 3 mg/m <sup>3</sup> | TWA: 3 mg/m <sup>3</sup> | TWA: 3 mg/m <sup>3</sup> | TWA: 3 mg/m <sup>3</sup> | (Vacated) TWA: 3.5 mg/m <sup>3</sup><br>TWA: 3.5 mg/m <sup>3</sup>                                                                                                         | IDLH: 1750 mg/m <sup>3</sup><br>TWA: 3.5 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH - NIOSH - National Institute for Occupational Safety and Health

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact,

and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

|                        |                                                                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| <b>Hand Protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                   |

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particle filter

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                          |
|-----------------------------------------------|--------------------------|
| <b>Physical State</b>                         | Solid                    |
| <b>Appearance</b>                             | Black                    |
| <b>Odor</b>                                   | Odorless                 |
| <b>Odor Threshold</b>                         | No information available |
| <b>pH</b>                                     | No information available |
| <b>Melting Point/Range</b>                    | No data available        |
| <b>Boiling Point/Range</b>                    | No information available |
| <b>Flash Point</b>                            | No information available |
| <b>Evaporation Rate</b>                       | Not applicable           |
| <b>Flammability (solid,gas)</b>               | No information available |
| <b>Flammability or explosive limits</b>       |                          |
| Upper                                         | No data available        |
| Lower                                         | No data available        |
| <b>Vapor Pressure</b>                         | <=1100 hPa @ 50 °C       |
| <b>Vapor Density</b>                          | Not applicable           |
| <b>Specific Gravity</b>                       | No information available |
| <b>Solubility</b>                             | No information available |
| <b>Partition coefficient; n-octanol/water</b> | No data available        |
| <b>Autoignition Temperature</b>               | No information available |
| <b>Decomposition Temperature</b>              | No information available |
| <b>Viscosity</b>                              | Not applicable           |

## 10. Stability and reactivity

|                                         |                                                                                                         |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                              |
| <b>Stability</b>                        | Stable under normal conditions.                                                                         |
| <b>Conditions to Avoid</b>              | Incompatible products.                                                                                  |
| <b>Incompatible Materials</b>           | Organic materials, Oxidizing agent                                                                      |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Sulfur oxides, Hydrogen fluoride, Metal oxides |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                           |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Mist LC50

Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.

#### Component Information

| Component    | LD50 Oral                  | LD50 Dermal              | LC50 Inhalation                           |
|--------------|----------------------------|--------------------------|-------------------------------------------|
| Graphite     | Not listed                 | Not listed               | LC50 > 2000 mg/m <sup>3</sup> ( Rat ) 4 h |
| Carbon black | LD50 > 15400 mg/kg ( Rat ) | LD50 > 3 g/kg ( Rabbit ) | LC50 > 4.6 mg/m <sup>3</sup> ( Rat ) 4 h  |

#### Toxicologically Synergistic Products

No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

#### Irritation

No information available

#### Sensitization

No information available

#### Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component               | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------|-----------|------------|------------|------------|------------|------------|
| Graphite                | 7782-42-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Polytetrafluoroethylene | 9002-84-0 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Carbon black            | 1333-86-4 | Group 2B   | Not listed | A3         | X          | A3         |

IARC (International Agency for Research on Cancer)

IARC (International Agency for Research on Cancer)

Group 1 - Carcinogenic to Humans

Group 2A - Probably Carcinogenic to Humans

Group 2B - Possibly Carcinogenic to Humans

A1 - Known Human Carcinogen

A2 - Suspected Human Carcinogen

A3 - Animal Carcinogen

ACGIH: (American Conference of Governmental Industrial Hygienists)

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#### Mutagenic Effects

No information available

#### Reproductive Effects

No information available.

#### Developmental Effects

No information available.

#### Teratogenicity

No information available.

#### STOT - single exposure

None known

|                                                   |                                                                |
|---------------------------------------------------|----------------------------------------------------------------|
| <b>STOT - repeated exposure</b>                   | None known                                                     |
| <b>Aspiration hazard</b>                          | No information available                                       |
| <b>Symptoms / effects, both acute and delayed</b> | No information available                                       |
| <b>Endocrine Disruptor Information</b>            | No information available                                       |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. |

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component    | Freshwater Algae | Freshwater Fish                                 | Microtox   | Water Flea                             |
|--------------|------------------|-------------------------------------------------|------------|----------------------------------------|
| Graphite     | Not listed       | LC50: > 100 mg/L, 96h semi-static (Danio rerio) | Not listed | Not listed                             |
| Carbon black | Not listed       | Not listed                                      | Not listed | EC50: > 5600 mg/L, 24h (Daphnia magna) |

**Persistence and Degradability** Insoluble in water May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

|                 |               |
|-----------------|---------------|
| <b>DOT</b>      | Not regulated |
| <b>TDG</b>      | Not regulated |
| <b>IATA</b>     | Not regulated |
| <b>IMDG/IMO</b> | Not regulated |

## 15. Regulatory information

### International Inventories

| Component               | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS    | NLP |
|-------------------------|-----------|-----|------|------|-----------------------------------------------|-----------|-----------|-----|
| Graphite                | 7782-42-5 | X   | -    | X    | ACTIVE                                        | 231-955-3 | -         | -   |
| Polytetrafluoroethylene | 9002-84-0 | X   | -    | X    | ACTIVE                                        | -         | -         | -   |
| Carbon black            | 1333-86-4 | X   | -    | X    | ACTIVE                                        | 215-609-9 | 435-640-3 | -   |

| Component               | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Graphite                | 7782-42-5 | X     | X        | -    | -    | X    | X    | X     | X     |
| Polytetrafluoroethylene | 9002-84-0 | X     | KE-33429 | X    | X    | X    | X    | X     | X     |
| Carbon black            | 1333-86-4 | X     | X        | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances  
 KECL - Korean Existing and Evaluated Chemical Substances  
 ENCS - Japanese Existing and New Chemical Substances  
 AICS - Australian Inventory of Chemical Substances  
 PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component    | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEMP)         |
|--------------|------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------|
| Carbon black |                                                      |                                                                            | Subject to Monitoring and Surveillance Activities |

## Other International Regulations

### Authorisation/Restrictions according to EU REACH

| Component    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Carbon black | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component               | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Graphite                | 7782-42-5 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Polytetrafluoroethylene | 9002-84-0 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Carbon black            | 1333-86-4 | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component               | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Graphite                | 7782-42-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Polytetrafluoroethylene | 9002-84-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Carbon black            | 1333-86-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

### Prepared By

Product Safety Department  
 Email: chem.techinfo@thermofisher.com  
 www.thermofisher.com

### Revision Date

02-April-2024

### Print Date

02-April-2024

### Revision Summary

New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,

transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**